1) Todos los polinomios de Bernoulli de grado 0 son iguales a 1: $\boxed{B_0^n(x)=1}$

$$\varphi(n;t;x) = \frac{t^n e^{tx}}{(e^t-1)^n} = \sum_{v=0}^{\infty} \frac{t^v}{v!} B_v^n(x)$$

$$t^n e^{tx} = (e^t-1)^n \cdot \sum_{v=0}^{\infty} \frac{t^v}{v!} B_v^n(x)$$

$$t^n \cdot \sum_{r=0}^{\infty} \frac{(tx)^r}{r!} = \left[\sum_{r=0}^{\infty} \frac{t^r}{r!} - 1 \right]^n \sum_{v=0}^{\infty} \frac{t^v}{v!} B_v^n(x)$$

$$t^n \left[1 + tx + (tx)^2 + \cdots \right] = \left[t + t^2 + \cdots \right]^n \left[B_0^n(x) + t B_1^n(x) + \cdots \right]$$

$$\left(t^n + t^{n+1} x + t^{n+2} x^2 + \cdots \right) = \left(t^n + \alpha t^{n+1} + \cdots \right) \cdot \left(B_0^n(x) + t B_1^n(x) + \cdots \right)$$

$$\left(t^n + t^{n+1} x + \cdots \right) = \left(t^n B_0^n(x) + t^{n+1} B_1^n(x) + \cdots \right) + \left(\alpha t^{n+1} + \cdots \right) \cdot \sum_{v=0}^{\infty} \frac{t^v}{v!} B_v^n(x)$$

luego $t^n = t^n B_0^n(x) \Rightarrow B_0^n(x) = 1$ q.e.d

2) Todos los polinomios de Bernoulli de orden 0 son polinomios mónicos elementales. $B_v^0(x) = x^v$

$$\varphi(0; t; x) = \frac{t^0 e^{tx}}{(e^t - 1)^0} = \sum_{v=0}^{\infty} \frac{t^v}{v!} B_v^0(x)$$

$$\sum_{r=0}^{\infty} \frac{(tx)^r}{r!} = \sum_{v=0}^{\infty} \frac{t^v}{v!} B_v^0(x)$$

$$\sum_{r=0}^{\infty} \frac{t^r}{r!} x^r = \sum_{v=0}^{\infty} \frac{t^v}{v!} B_v^0(x)$$

Luego: $x^v = B_v^0(x) \quad r \in \mathbb{N}_0$

3) Derivada de un polinomio de Bernoulli: $\boxed{D B_v^n(x) = v B_{v-1}^n(x)}$

$$\varphi(n;\tau;x) = \frac{\tau^n e^{\tau x}}{(e^{\tau}-1)^n} = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_v^n(x)$$

Deriw respecto a x:

$$D_x\left[\frac{\tau^n e^{\tau x}}{(e^{\tau}-1)^n}\right] = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} D\left(B_v^n(x)\right)$$

$$\left(\frac{\tau^n}{(e^{\tau}-1)^n}\, e^{\tau x}\right)\tau = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} D\left(B_v^n(x)\right)$$

$$\tau \cdot \sum_{v=0}^{\infty} \frac{\tau^v}{v!} B_v^n(x) = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} D\left(B_v^n(x)\right)$$

$$\sum_{v=0}^{\infty} \frac{\tau^{v+1}}{(v+1)!} \cdot (v+1)\, B_v^n(x) = \sum_{v=0}^{\infty} \frac{\tau^v}{v!} D\left(B_v^n(x)\right)$$

$$\sum_{v=1}^{\infty} \frac{\tau^v}{v!} \cdot v\, B_{v-1}^n(x) = \underbrace{\frac{\tau^0}{0!} D\underbrace{\left(\overbrace{B_0^n(x)}^{=1}\right)}_{=0}}_{=0} + \sum_{v=1}^{\infty} \frac{\tau^v}{v!} D\left(B_v^n(x)\right)$$

$$\sum_{v=1}^{\infty} \frac{\tau^v}{v!} \cdot \underline{v\, B_{v-1}^n(x)} = \sum_{v=1}^{\infty} \frac{\tau^v}{v!} D\left[\underline{B_v^n(x)}\right]$$

$$\boxed{\text{Luego } \quad v B_{v-1}^n(x) = D\left[B_v^n(x)\right]}$$

4) Diferencia de un polinomio de Bernoulli. $\boxed{\Delta_{h=1} B_v^n(x) = v\, B_{v-1}^{n-1}(x)}$

$$\Delta_{h=1}\left[\frac{t^n e^{tx}}{(e^t-1)^n}\right] = \sum_{v=0}^{\infty} \frac{t^v}{v!}\,\Delta\left(B_v^n(x)\right)$$

$$\frac{t^n}{(e^t-1)^n}\left(e^{t(x+1)} - e^{tx}\right) = \sum_{v=0}^{\infty} \frac{t^v}{v!}\,\Delta\left(B_v^n(x)\right)$$

$$\frac{t^n e^{tx}}{(e^t-1)^n}\left(e^t-1\right) = \sum_{v=0}^{\infty} \frac{t^v}{v!}\,\Delta\left(B_v^n(x)\right)$$

$$t\cdot \frac{t^{n-1} e^{tx}}{(e^t-1)^{n-1}} = \sum_{v=0}^{\infty} \frac{t^v}{v!}\,\Delta\left(B_v^n(x)\right)$$

$$t\cdot \sum_{v=0}^{\infty} \frac{t^v}{v!}\, B_v^{n-1}(x) = \sum_{v=0}^{\infty} \frac{t^v}{v!}\,\Delta\left(B_v^n(x)\right)$$

$$\sum_{v=0}^{\infty} \frac{t^{v+1}}{(v+1)!}(v+1)\, B_v^{n-1}(x) = \frac{t^0}{0!}\underbrace{\Delta\big(\overbrace{B_0^n(x)}^{=1}\big)}_{=0} + \sum_{v=1}^{\infty} \frac{t^v}{v!}\,\Delta\left(B_v^n(x)\right)$$

$$\sum_{v=1}^{\infty} \frac{t^v}{v!}\cdot v\, B_{v-1}^{n-1}(x) = 0 + \sum_{v=1}^{\infty} \frac{t^v}{v!}\,\Delta\left(B_v^n(x)\right)$$

Luego $\boxed{v\, B_{v-1}^{n-1}(x) = \Delta B_v^n(x)}$